FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

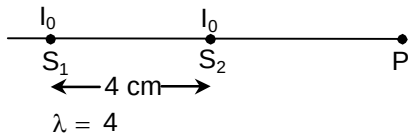
PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Two coherent sources S_1 and S_2 are situated as shown in figure. Here S_1 is fixed but S_2 is performing SHM with amplitude 2 cm. Initially S_2 is at right extreme. At time $t = \frac{\pi}{2\omega}$ (where ω = angular frequency of SHM):



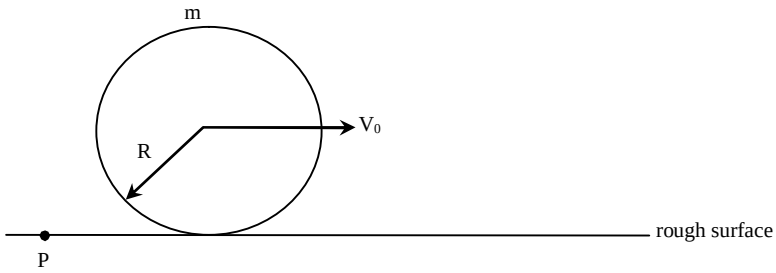
(A) Phase difference between waves produce by S_1 and S_2 at point P is $\frac{\pi}{2}$.

(B) Intensity at point P is $7I_0$.

(C) Intensity at point P is $4I_0$

(D) Intensity at point P is zero

2. A disc of mass m and radius R is projected with initially velocity V_0 and initial angular velocity zero on rough surface. Then, which of the following is **INCORRECT**.



(A) Angular momentum of disc about point P is conserved

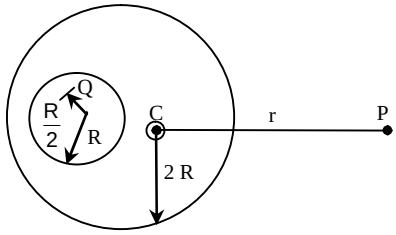
(B) Friction force will be zero after pure rolling start

(C) Final angular velocity of disc will be $\frac{V_0}{R}$

(D) Final velocity of centre of mass will be $\frac{2V_0}{3}$

Space for rough work

3. There is a conductor with cavity of radius R as shown in figure. A point charge Q is placed at R/2 distance from centre of cavity. A point P is at r distance from centre of conductor. Then, which of the following is **INCORRECT**.



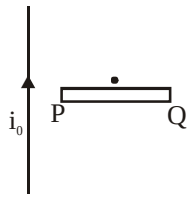
(A) charge density at inner surface of cavity will be non uniform

(B) charge density at surface of conductor will be uniform

(C) electric field at point P is $\frac{Q}{4\pi\varepsilon_0 r^2}$

(D) force on charge Q is zero.

4. A metal rod PQ (carrying current from P to Q) is placed perpendicular to the infinite long wire carrying current i_0. If this arrangement lies in a horizontal plane; then the rod PQ will rotate:

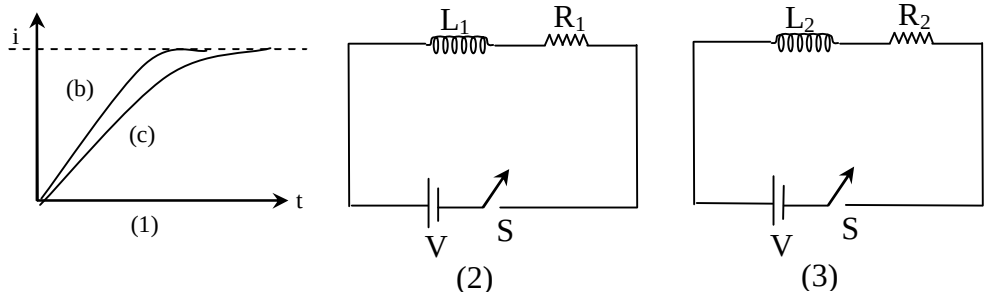


(A) clockwise (B) anticlockwise

(C) along PQ (D) does not rotate

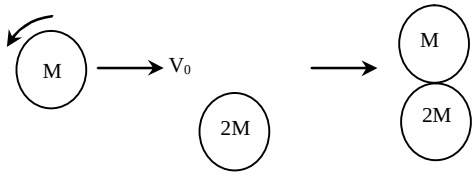
Space for rough work

5. Current growth in two L – R circuits (2) and (3) is shown in figure (1) after switch S closed. Then



(A) $R_1 > R_2$
(B) $R_1 = R_2$
(C) $L_1 > L_2$
(D) $L_1 = L_2$

6. A uniform disc of mass m and diameter 2 R moves forward towards another uniform disc of mass 2 m & diameter 2 R on a frictionless surface as shown in figure



When the first disc contacts the second, they stick to each other and move as a single object. Then, find the **incorrect** option.

(A) the velocity of combined disc after the collision is $\frac{V_0}{3}$.

(B) the angular velocity of combined disc after the collision is $\frac{\omega_0}{3}$.

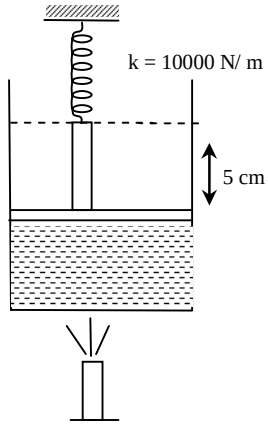
(C) If $\omega_0 = \frac{8}{3}\frac{V_0}{R}$, the combined disc will not rotate

(D) If combined disc does not rotate, the energy loss is $\frac{19}{9}MV_0^2$.

Space for rough work

7. Two strings with linear mass densities $\mu_1 = 0.1\,\text{kg/m}$ and $\mu_2 = 0.3\,\text{kg/m}$ are joined seamlessly. They are under tension of 20 N. A travelling wave of triangular shape is moving from lighter to heavier string.
   (A) The reflection coefficient at interface is zero
   (B) The reflection coefficient at interface is $2-\sqrt{3}$
   (C) The transmission coefficient at interface is 1
   (D) The transmission coefficient at interface is $\sqrt{3}+5$

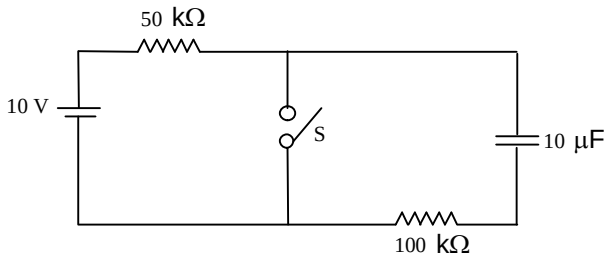
8. Consider a piston – cylinder arrangement with spring in its natural length. Due to heat transfer, the gas expands until the piston hits the stop. The mass of piston is 10 kg and area is 78 cm^2. Initial and final specific internal energies are 214 and 337 k J / kg. Then, find the **incorrect** option.
   (A) Initial pressure of gas is 112 k Pa
   (B) Work done by gas on the piston is 56.5 J
   (C) Work done by gas on the piston is 100 J
   (D) Heat transfer for 0.5 gas is 118 J



Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

9. In the circuit shown, the switch S has been closed for a long time. At time t = 0, the switch is opened. It remains open for a long time T, after which it is closed again.



   (A) voltage drop across 100 kΩ resister is $10e^{-t/1.5}$ V for t < T
   (B) voltage drop across 100 kΩ resister is $\frac{20\,e^{-t/1.5}}{3}$ V for t < T
   (C) voltage drop across 100 kΩ resister is $100\left(e^{T-t}-e^{(T-3t)/3}\right)$ for t > T
   (D) The time constant of the circuit is 1.5 sec for t > T

Space for rough work

10. Which type of electrostatic field $\bar{E}$ is not possible?
(A) $\bar{E} = y\hat{i}$ (B) $\bar{E} = x\hat{j}$
(C) $\bar{E} = y\hat{i} + x\hat{j}$ (D) $x^2\left(3y\hat{i} + x\hat{j}\right)$

11. The threshold wavelength for photoelectric emission from a material is $5200\,\text{Å}$, photoelectrons will be emitted when this material is illuminated with monochromatic radiation from a
(A) 50 W infrared lamp (B) 100 W infrared lamp
(C) 50 W ultraviolet lamp (D) 100 W ultraviolet lamp

12. A very large number of fine smoke particles each of mass m_1 are trapped inside a small glass container, which contains an ideal gas of molecular mass m_2 at constant pressure P and density d. On viewing the particles through a powerful microscope, the ratio of the rms speed of the smoke particles to that of the gas molecules is K.
(A) smoke particles are in random translatory motion
(B) $m_1 = K^2 m_2$
(C) $m_2 = K^2 m_1$
(D) rms speed of the smoke particles will change if (p : d) changes.

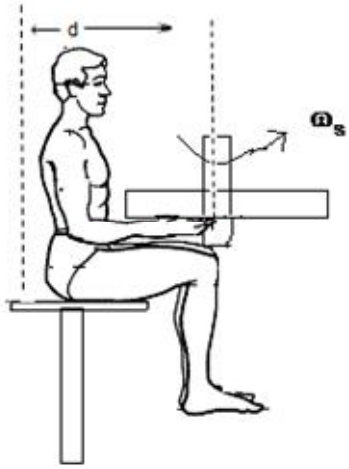
Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Questions 13 & 14

A person is sitting on a stool that is initially not rotating and is holding a spinning wheel. The moment of inertia of the person and stool about a vertical axis passing through the centre of stool is I_{sp}. The moment of inertia of the wheel about an axis, perpendicular to the plane of the wheel, passing through the centre of mass of the wheel is $I\omega = \left(\frac{1}{4}\right)I_{SP}$. The mass of wheels m_ω. Suppose that the person holds the wheel as shown in the sketch such that the distance of an axis passing through the centre of mass of wheel to the axis of rotation of stool is d and $m_\omega d^2 = \frac{1}{3}I_\omega$. Suppose the wheel is spinning initially at an angular velocity ω_s. The person then turns the spinning wheel upside down. You may ignore any frictional torque in the bearing of the stool.



13. What is angular speed of person and stool after the spinning wheel is turned upside down?

(A) $\frac{3}{4}w_s$ (B) $\frac{3}{2}w_s$

(C) $\frac{3}{8}w_s$ (D) $\frac{3}{7}w_s$

14. Linear impulse given by person to wheel.

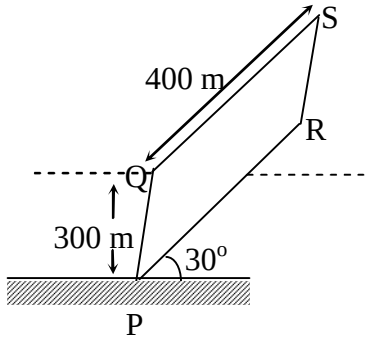
(A) $\frac{3}{4}m_\omega dw_s$ (B) $\frac{3}{2}m_\omega dw_s$

(C) $\frac{3}{8}m_\omega dw_s$ (D) $\frac{3}{7}m_\omega dw_s$

Space for rough work

Paragraph for Questions 15 & 16

A rectangular plank of sides 300 m x 400 m is placed over a horizontal plane at an inclination of 30° as shown in the figure. (Take g = 10 m/s^2)



15. A particle is projected from one base corner at an angle of 30° with the plank but in a vertical plane which carries line of maximum possible length of plank with speed 10 m/s. Then, if time after which particle strikes the plank is T sec, find $T\sqrt{21}$.

(A) 5 sec
(B) $2\sqrt{7}$ sec
(C) $\frac{2}{\sqrt{3}}$ sec
(D) None

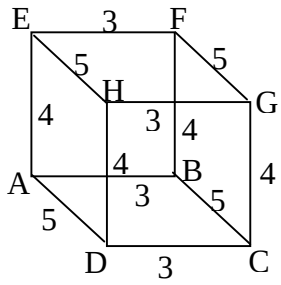
16. If a particle was projected with speed 10 m/s from one base corner at an angle 30° with the plank, but in a vertical plane which carries a line over the plank with least possible slope, then time after which particle lands over the plank is

(A) 1 sec
(B) $\frac{2}{\sqrt{3}}$ sec
(C) 2 sec
(D) Never

Space for rough work

Paragraph for Questions 17 & 18

An insect which can crawl only, but can not fly, has to reach from one corner to diagonally opposite corner inside a room ABCDEFGH.



17. Find the minimum path length which the insect may select to reach the target point [AD = 5 m, DH = 4 m, DC = 3 m]

(A) 10 m (B) $\sqrt{90}$ m

(C) $5\sqrt{2}$ m (D) $\sqrt{74}$ m

18. If while crawling over wall velocity of insect is 2 m/hr and over the floor it is 4 m/hr, then find the minimum possible time insect may take to reach the target point.

(A) More than 3.37 hours (B) 3.37 hours

(C) Less then 3.37 hours (D) Less than 1 hour

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

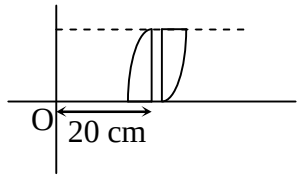
1. If two spherical bubbles of same radius 2 m. and same liquid coalesece to form a common plane surface, then find the gap between the centres of the bubbles.

2. If the no. of times the electron goes round the first Bohr orbit of hydrogen atom in 1S is $P \times 10^{15}$ per second then find approximate value of $\frac{2P}{13}$.

3. A uniformly charged non conducting disc of radius 2 m, mass 1 kg and charges Q = 4C is lying over smooth horizontal plane. The space carries a uniform vertical magnetic field of 1 Tesla find the angular velocity of disc just after reversal in direction of magnetic field.

4. Sound wave travels through air between two points A & B located at a gap of 30 m. if temperature between them varies linearly from -173^0 C to 127^oC. Time taken by sound signal to move from A to B is $K\sqrt{\frac{M}{\gamma R}}$ where γ is adiabatic exponent of air and M its molecular weight. Then find value of K.

5. A biconvex thin lens of focal length 30 cm and aperture diameter 10 cm is broken into four identical pieces. Two of the pieces are placed in a coordinate system as shown in the diagram. Find the magnitude of the y coordinate of the image if an object is placed at origin.



Space for rough work

Chemistry

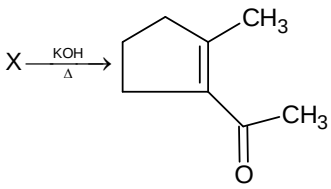
PART – II

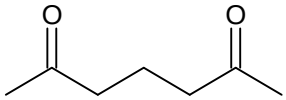
SECTION – A

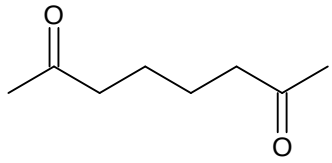
Straight Objective Type

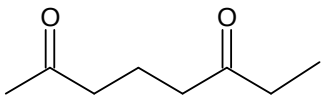
This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Write the order of reactivity of halogens towards electrophilic addition reactions of alkenes through cyclohalonium intermediate:
   (A) $F_2 > Cl_2 > Br_2 > I_2$
   (B) $I_2 > Br_2 > Cl_2 > F_2$
   (C) $I_2 > Br_2 > F_2 > Cl_2$
   (D) $Br_2 > Cl_2 > I_2 > F_2$

2. 

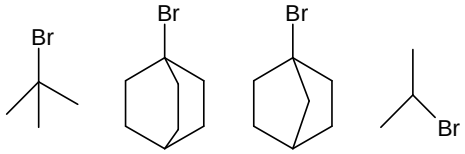
   Find out the structure of 'X'
   (A) 

   (B) 

   (C) 

   (D) None of these

3. Which is the correct order of following compounds towards nucleophilc substitution via 1st order mechanism?



(I) (II) (III) (IV)

   (A) I > IV > II > III
   (B) I > II > III > IV
   (C) I > IV > III > II
   (D) II > I > III > IV

Space for rough work

4. Determine which reagent is in excess and by how much if 100 g P_4O_6 is treated with 100g $KMnO_4$ in HCl solution to form H_3PO_4 and $MnCl_2$?
(A) P_4O_6, 12.05g (B) $KMnO_4$, 12.05g
(C) P_4O_6, 13.01g (D) $KMnO_4$, 13.01g

5. The equilibrium pressure of $NH_4CN(s) \rightleftharpoons NH_3(g) + HCN(g)$ is 0.298 atm. Calculate K_P. If $NH_4CN(s)$ is allowed to decompose in presence of NH_3 at 0.25 atm. Calculate partial pressure of HCN at equilibrium.
(A) $K_P = 0.022\ atm^2$, 0.0694 atm (B) $K_P = 0.022\ atm^2$, 0.0492 atm
(C) $K_P = 0.022\ atm^2$, 0.0542 atm (D) $K_P = 0.022\ atm^2$, 0.0601 atm

6. $AgBr(s) + 2S_2O_3^{2-}(aq.) \rightleftharpoons Ag(S_2O_3)_2^{3-}(aq) + Br^-(aq.)$
Given $K_{sp(AgBr)} = 5 \times 10^{-13}$, $K_{f\left(Ag(S_2O_3)_2^{3-}\right)} = 5 \times 10^{13}$
What is the molar solubility of AgBr in 0.1 M $Na_2S_2O_3$?
(A) 0.5 M (B) 0.45 M
(C) 0.045 M (D) None of these

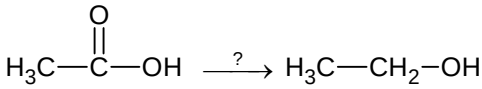
7. Energy of d-orbitals in square pyramidal geometry
(A) $d_{xz} \simeq d_{yz} < d_{z^2} < d_{xy} < d_{x^2-y^2}$ (B) $d_{xz} \simeq d_{yz} < d_{xy} < d_{z^2} < d_{x^2-y^2}$
(C) $d_{xz} < d_{yz} \simeq d_{xy} < d_{x^2-y^2} < d_{z^2}$ (D) $d_{xz} < d_{yz} \simeq d_{xy} < d_{x^2-y^2} < d_{z^2}$

8. When alkali metals dissolved in liq. NH_3 it turns blue colour solution and after some time it disappears. Then
(A) The solution acts as good conductor.
(B) The solution shows diamagnetic nature.
(C) The solution acts as strong oxidising agent.
(D) The solution liberates H_2 gas.

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

9. 

For above reaction, the reagent used is:
(A) $LiAlH_4$ (B) $NaBH_4$
(C) B_2H_6 (D) Ni/H_2

10. Which of the following possess two lone pairs at central atom and has square planar shape?
(A) SF_4 (B) XeO_4
(C) XeF_4 (D) ICl_4^-

11. In co-ordination compounds, the valence bond theory does not explain:
(A) Magnetic moment (B) Colour property
(C) Square planar geometry of Cu^{2+} complex (D) Square planar geometry of Pt^{2+}, Pd^{2+}

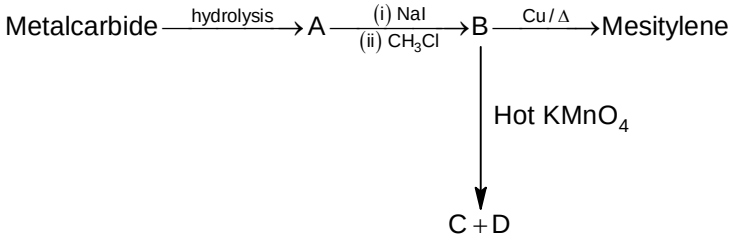
12. 1 atm pressure is equal to:
(A) 760 torr (B) 1.01325×10^5 pascal
(C) 1.01325×10^5 N/m^2 (D) 1.01325 bar

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 13 to 14



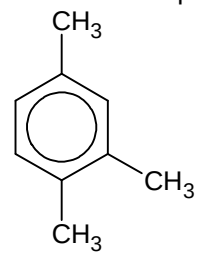
'C' gives effervescence with $NaHCO_3$

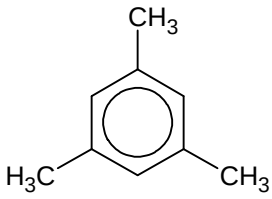
13. 'B' is

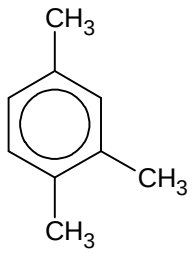
(A) $HC\equiv CH$ (B) $H_3C-CH_2-C\equiv C-H$

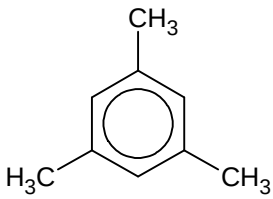
(C) $H_3C-C\equiv\equiv C-H$ (D) $H_3C-C\equiv\equiv C-CH_3$

14. Mesitylene and 'D' respectively are:

(A)  , CH_3COOH

(B)  , CO_2

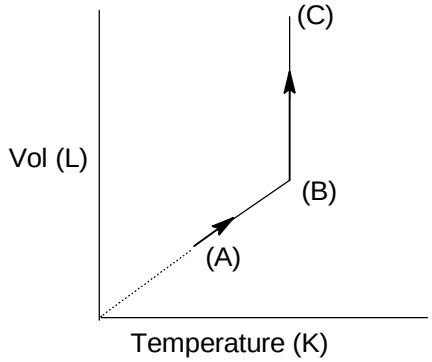
(C)  , CO_2

(D)  , HCOOH

Space for rough work

Paragraph for Question Nos. 15 to 16

A triatomic linear (neglect vibration degree of freedom) ideal gas of two mole is taken through a reversible process starting from 'A' as shown in figure



The volume ratio $\frac{V_B}{V_A} = 4$ and if the temperature at A is –73°C

15. Work done by the gas in AB process is
(A) 3.32 kJ (B) - 3.32 kJ
(C) 9.97 kJ (D) - 9.97 kJ

16. Total enthalpy change in the process (A $\rightarrow$C) is:
(A) 3000 R (B) 4200 R
(C) 2100 R (D) 0

Space for rough work

Paragraph for Question Nos. 17 to 18

Many bond angles can be explained by either electronegativity (or) size arguments. Molecules with larger difference in electronegativity values between central and outer atoms have smaller bond angles.

17. Which has the smallest bond angle (X—O—X) in the given molecules?
(A) OSF_2 (B) $OSCl_2$
(C) $OSBr_2$ (D) OSI_2

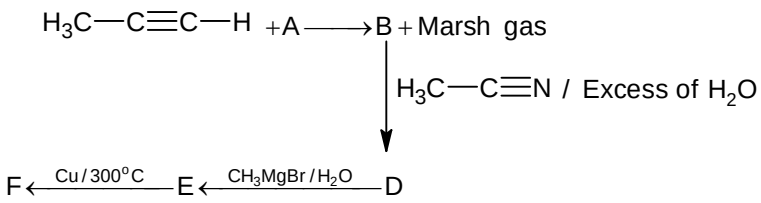
18. Consider the following iodides:
PI_3 AsI_3 SbI_3
102° 100.2^o 99°
The bond angle is maximum in PI_3 which is
(A) due to small size of phosphorus (B) due to more bp–bp repulsion in PI_3
(C) due to less electronegativity of 'P' (D) None of the above

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. 

No. of π bonds present in F

2. $C_6H_{14}O$ is an 3° alcohol (A) which on Cu/300°C gives an alkene (having four hyperconjugative structures) which reacts with NBS in presence of light gives Bromo substituted compound (B). B again reacts with Benzene in presence of $AlCl_3$ gives final product C. 'C' reacts with strong oxidising agent such as $KMnO_4$ to give CO_2 as one of the products. What is the number of 1° carbons present in (A).

3. Ammonium nitrate on strong heating, is decomposed to give a gas (A) and water. It is neutral gas. (A) on heating at 1300°C gives two gases (B) and (C). (B) reacts with $FeSO_4$ solution to form brown coloured compound (D). Number of unpaired electrons present in (D):

4. In Goldschmidt aluminothermic process, thermite mixture contains how many parts of Fe_2O_3 along with one part of aluminium?

5. At 48°C, the vapour pressure of pure CS_2 is 850 torr. A solution of 2.0 g of sulphur in 100 g of CS_2 has a vapour pressure 844.9 torr. Determine the atomicity of sulphur molecule.

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Let a, b and c be non-zero real numbers, such that $\int_0^1 (1+\cos^8 x)(ax^2+bx+c)dx = \int_0^2 (1+\cos^8 x)(ax^2+bx+c)dx$ then, the equation $ax^2 + bx + c = 0$ has
   (A) no roots in (0, 2)
   (B) atleast one root in (1, 2)
   (C) a double root in (0, 2)
   (D) two imaginary roots

2. The value of b, such that the equation $\frac{b\cos x}{2\cos 2x - 1} = \frac{b+\sin x}{(\cos^2 x - 3\sin^2 x)\tan x}$ posses solutions, belong to the set
   (A) $\left(-\infty, \frac{1}{2}\right)$
   (B) $\left(\frac{1}{2}, \infty\right)$
   (C) $(-\infty, \infty)$
   (D) $\left(-\infty, \frac{1}{2}\right) \cup (1, \infty)$

3. Two forces whose magnitudes are 2 gm wt and 3 gm wt act on a particle in the directions of the vectors $2\hat{i} + 4\hat{j} + 4\hat{k}$ and $4\hat{i} - 4\hat{j} + 2\hat{k}$ respectively. If the particle is displaced from the origin to the point (1, 2, 2), then work done is
   (A) 6 gm-cm
   (B) 4 gm-cm
   (C) 5 gm-cm
   (D) none of these

Space for rough work

4. If a, b, c, d be in G.P. then $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2)$ is
(A) $ab + bc + cd$ (B) $(ab + bc + cd)^2$
(C) $(ab - bc - cd)^2$ (D) 0

5. The value of 'a' for which $ax^2 + \sin^{-1}(x^2 - 2x + 2) + \cos^{-1}(x^2 - 2x + 2) = 0$ has a real solution, is
(A) $\frac{\pi}{2}$ (B) $-\frac{\pi}{2}$
(C) $\frac{2}{\pi}$ (D) $-\frac{2}{\pi}$

6. Let $f(x) = \begin{cases} x^p \sin\left(\frac{1}{x}\right) & , \ x \neq 0 \\ 0 & , \ x = 0 \end{cases}$, then f(x) is continuous but not differentiable at x = 0 if
(A) $p < 0$ (B) $p = 0$
(C) $0 < p \leq 1$ (D) $p \geq 1$

7. If $A_1, A_3, \ldots A_{2n-1}$ are n skew–symmetric matrices of same order then $B = \sum_{r=1}^{n} (2r-1)\left(A_{2r-1}\right)^{2r-1}$ will be
(A) symmetric (B) skew–symmetric
(C) neither symmetric nor skew–symmetric (D) data is inadequate

8. In an acute angled triangle ABC, vertex A is (1, 2). Let H is the orthocentre and M is the mid point of side BC. On the line HM, a point T is taken such that HM = MT and equation of lines BT and CT are $x - 2y + 3 = 0$ and $x + y + 1 = 0$ respectively. Find equation of circumcircle of ΔABC.
(A) $(x+1)\left(x+\frac{5}{8}\right)+(y-2)\left(y-\frac{2}{3}\right)=0$ (B) $(x+1)\left(x+\frac{5}{8}\right)+(y+2)\left(y+\frac{2}{3}\right)=0$
(C) $(x-1)\left(x-\frac{5}{8}\right)-(y-2)\left(y-\frac{2}{3}\right)=0$ (D) $(x-1)\left(x+\frac{5}{8}\right)+(y-2)\left(y-\frac{2}{3}\right)=0$

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

9. Let $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)=\alpha$, $\tan^{-1}\left(-\tan\frac{2\pi}{3}\right)=\beta$. Then

(A) $\alpha > \beta$ (B) $4\alpha - 3\beta = 0$

(C) $\alpha+\beta=\frac{7\pi}{12}$ (D) $\alpha + \beta = \pi$

10. If $f(x)=\left|\left(x+\frac{1}{2}\right)[x]\right|$, $x \in [-2, 2]$ (where [.] denotes the G.I.F.), then f(x) is discontinuous at x is equal to

(A) -2 (B) 2

(C) 0 (D) 1

11. If D_1 and D_2 are diagonal matrices of order 3, then

(A) D_1D_2 is a diagonal matrix (B) $D_1D_2 = D_2D_1$

(C) $D_1^2+D_2^2$ is a diagonal matrix (D) none of these

12. Let $u_n=\int_0^{\pi/2}\cos^n x\cos nx\,dx$ it turns out that u_1, u_2, u_3, ... form a G.P. then

(A) the common ratio of G.P. is $\frac{1}{2}$ (B) the common ratio of G.P. is $\frac{1}{4}$

(C) then $u_n=\frac{\pi}{2^{2n}}$ (D) then $u_n=\frac{\pi}{2^{n+1}}$

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:
If the three equation $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a + b)x + 36 = 0$ have a common possible root

13. Then value of a and b are
(A) $a = \pm 7, b = \pm 8$ (B) $a = \pm 8, b = \pm 7$
(C) $a = 0, b = 0$ (D) none of these

14. Then the common possible roots are
(A) ± 1 (B) ± 2
(C) ± 3 (D) ± 4

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

P: $y^2 = 8x$ and E: $\frac{x^2}{4} + \frac{y^2}{15} = 1$ are equation of parabola and ellipse respectively

15. Equation of a tangent common to both the parabola P and ellipse E is
(A) $x - 2y + 8 = 0$ (B) $x + y + 9 = 0$
(C) $x + 2y - 8 = 0$ (D) $x - 2y - 8 = 0$

16. Equation of the normal at the point of contact of the common tangent, which (i.e. tangent) makes an acute angle with positive direction of x-axis to the parabola P, is
(A) $2x + y = 24$ (B) $2x + y + 24 = 0$
(C) $2x + y = 48$ (D) $2x + y + 48 = 0$

Space for rough work

Paragraph for Question Nos. 17 to 18

Read the following write up carefully and answer the following questions:
Let $f(x) = |\ln|\{x\} - 1||\ \forall\ x \in [2, 3)$ and $g(x) = f'(x)$. If $h'(x) = f^2(x) + g^2(x)$ such that $h(2) = -3$, then (where $\{x\}$ denotes fractional part of x)

17. $h(x) + g(x) = 0$ has
(A) at least one root in [2, 3) (B) exactly one root in [2, 3)
(C) non real root in [2, 3) (D) can't say

18. $h'(x) + g'(x) = 0$ has
(A) at least one root in [2, 3) (B) exactly one root in [2, 3)
(C) no real root in [2, 3) (D) can't say

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. If $\int \frac{\cos x(1+4\cos 2x)}{\sin x + 4\sin x\cos^2 x}dx = \frac{1}{2}\ln|h(x)| + \frac{1}{2}\ln|f(x)| + c$ then, $\left|f\left(\frac{\pi}{2}\right)\cdot h\left(\frac{\pi}{2}\right)\right|$ is equal to _____

2. If $\int_0^{x^2} f(t)dt = x\cos\pi x$ and $\int_0^{f(x)} t^2dt = x\cos\pi x$ then $\frac{f'(4)}{16}$ is equal to _____

3. Let f: R $\to$ R be a twice differentiable function satisfying $f''(x) - 5f'(x) + 6f(x) \geq 0\ \forall\ x \geq 0$ if $f(0) = 1$, $f'(0) = 0$. If f(x) satisfies $f(x) \geq ah(bx) - bh(ax)$; $\forall\ x \geq 0$, then the value of $(a + b)\ h(0)$ is equal to _____

4. Let f(x) be a differentiable function such that $f'(x) + f(x) = 4xe^{-x} \cdot \sin 2x$ and $f(0) = 0$ if $\lim_{n\to\infty}\sum_{k=1}^{n} f(k\pi) = \frac{-p\pi e^{\pi}}{\left(e^{\pi}-1\right)^2}$ then p is equal to _____

5. If the range of the function $\cos^{-1}|1-x^2|$ is $\left[a, \frac{\pi}{b}\right]$ then find the value of b _____

Space for rough work